WORKSHEET 6.5

Analysing a Function Defined by an Integral

Unit 6: Integration and Accumulation of Change

Key ideas

- If $g(x) = \int_a^x f(t)\,dt$, then $g'(x) = f(x)$ and $g''(x) = f'(x)$.
- g is increasing where f is positive; g has a relative maximum where f changes from positive to negative.
- The graph of g is concave up where f is increasing, and has a point of inflection where f changes direction.
- Values of g are signed areas under the graph of f.

Worked examples

Example 1 · A relative minimum

Let $g(x) = \int_0^x (t-3)\,dt$. Find the relative minimum of g.

Solution. $g'(x) = x - 3$ changes from negative to positive at $x = 3$. $g(3)$ is the signed area of a triangle below the axis.

Relative minimum $g(3) = -\frac{9}{2}$.

Example 2 · Concavity

If $g' = f$ and f is decreasing on $(2, 5)$, what is true of the graph of g there?

Solution. $g'' = f' < 0$.

It is concave down.

Example 3 · An absolute extremum

Let $g(x) = \int_0^x \left(4 - t^2\right)\,dt$. Find the absolute maximum of g on $[0, 3]$.

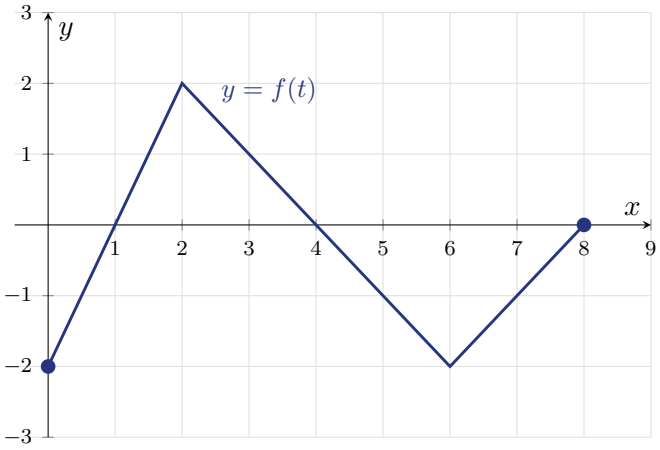
Solution. $g'(x) = 4 - x^2 = 0$ at $x = 2$. $g(0) = 0$, $g(2) = 8 - \frac{8}{3} = \frac{16}{3}$, $g(3) = 12 - 9 = 3$.

$\frac{16}{3}$**, at** $x = 2$**.**

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Practice

Questions 1 to 8 use this graph of f, which consists of three line segments. Let $g(x) = \int_0^x f(t)\,dt$.



CORE

1. Find $g(2)$.

2. Find $g'(2)$.

3. Find $g(4)$.

4. Find $g''(3)$.

EXAM LEVEL

5. Find the x-coordinates of the relative minimum and the relative maximum of g on $(0, 8)$. Justify your answers.

6. Find the absolute maximum and absolute minimum values of g on $[0, 8]$.

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7. On what intervals is the graph of g concave up? Where are its points of inflection?

8. Write an equation of the line tangent to the graph of g at $x = 6$.

Question 9 · Free-response practice

Let $H(x) = \int_1^x \ln t \, dt$ for $x > 0$.

(a) Find $H(1)$ and $H'(x)$.

(b) Show that H has its absolute minimum at $x = 1$.

(c) Describe the concavity of the graph of H. Given that $H(e) = 1$, write an equation of the tangent line at $x = e$.

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Worked solutions

1. The areas on $[0, 1]$ and $[1, 2]$ are -1 and 1: $g(2) = 0$.
2. $g'(2) = f(2) = 2$.
3. $g(2) + 2 = 2$ (a triangle of area 2 on $[2, 4]$).
4. The slope of f on $(2, 6)$: -1.
5. Minimum at $x = 1$, where $g' = f$ changes from negative to positive. Maximum at $x = 4$, where f changes from positive to negative.
6. $g(0) = 0$, $g(1) = -1$, $g(4) = 2$, $g(8) = 2 - 2 - 2 = -2$. Maximum 2; minimum -2 (at $x = 8$).
7. Concave up on $(0, 2)$ and $(6, 8)$, where f is increasing. Points of inflection at $x = 2$ and $x = 6$.
8. $g(6) = 0$ and $g'(6) = f(6) = -2$: $y = -2(x - 6)$.
9. (a) $H(1) = 0$ and $H'(x) = \ln x$.
   (b) $H'(x) < 0$ for $0 < x < 1$ and $H'(x) > 0$ for $x > 1$, so H decreases and then increases: the absolute minimum is $H(1) = 0$.
   (c) $H''(x) = \dfrac{1}{x} > 0$: concave up for all $x > 0$. The slope at $x = e$ is $\ln e = 1$: $y - 1 = x - e$.

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